# Practical — Are You Rational?

Advanced Microeconomics 1 · Thursday 1 October, 11:00–13:00 · bring a laptop with R

## What this session is

In the Session 1 quiz you divided 100 tokens between two accounts, one paying euros if a single coin flip lands heads and one if tails, twenty-five times at varying rates. Each decision was a point on a budget line: writing $x_1$ for the heads-claim and $x_2$ for the tails-claim, you chose a bundle with $p_1x_1 + p_2x_2 = 100$, at a different price vector each round.

In this session you will write, from a blank R script, the complete test of whether those 25 choices are consistent with maximizing *any* well-behaved utility function — and then quantify how far they fall short. No packages, no template code: everything is built from loops you write yourself, checked against a small example whose answer you are told. (The same test has been run on lab subjects, on a representative Dutch panel, and recently on GPT — Chen et al. 2023. Today it runs on you.)

## Before you arrive (this is the whole preparation)

A laptop with a **working R installation** (RStudio or plain R, either is fine). Confirm it works by running two lines — any two lines. If R starts and `1 + 1` answers, you are ready. The class dataset `rounds.csv` is distributed at the start of the session; find your rows with your quiz ID.

## The session, step by step

### Part 1 — Look at the data (~15 min)

Load `rounds.csv` with `read.csv`; pull out your own 25 rows; verify $p_1x_1 + p_2x_2 = 100$ in every round (it will be — ask yourself why that makes it *untestable*); plot your 25 budget lines with your 25 chosen points on them.

### Part 2 — Build the rationality test (~50 min)

Do not start on the real data. Start on three made-up rounds where the answer is known — the same numbers you will work *by hand* in Exercise Class 1:

| round | $p_1$ | $p_2$ | $x_1$ | $x_2$ |
|---|---|---|---|---|
| 1 | 2 | 1 | 3 | 2 |
| 2 | 1 | 2 | 2 | 3 |
| 3 | 1 | 1 | 4 | 1 |

The steps, each a few lines of loops:

1. **The cost table.** Build the $T \times T$ matrix $E$ with $E[t, s] = p_1^t x_1^s + p_2^t x_2^s$: what bundle $s$ would have cost at round $t$'s prices. Two nested `for` loops. The diagonal is what was actually spent.
2. **Revealed preference.** Two TRUE/FALSE matrices: `RD[t,s]` if $E[t, t] \geq E[t, s]$ (bundle $s$ was affordable when $t$ was chosen: $t$ revealed at least as good as $s$) and `PD[t,s]` for the strict version. Add a tolerance of `1e-9` to every comparison.
3. **Chains.** Preference chains through intermediate rounds: extend `RD` by allowing each round $k$ to act as a stepping stone — a third loop wrapped around the two you have (`if RC[t,k] and RC[k,s] then RC[t,s] <- TRUE`).
4. **Violations.** Count pairs where the chain says $t \succsim s$ while the data say $s$ strictly beats $t$: `RC[t,s] && PD[s,t]`.

**Checkpoint: the toy data produce exactly 3 violations.** Then overwrite the four input vectors with your own data ($T = 25$ now) and re-run the identical loops. The loops do not care.

### Part 3 — The CCEI (˜30 min)

Violation counts are crude. Afriat's question: *how much of your budget must be wasted before your choices become coherent?* Pretend the person had only a fraction $e$ of what they spent: replace $E[t, t]$ by $e \cdot E[t, t]$ in step 2 — one new symbol. Then search: start at $e = 1$ and walk down in steps of 0.005, re-running your whole test, until the violations vanish. The first $e$ that works is your **CCEI**; $1-$ CCEI is the share of your budget your inconsistency burns.

**Checkpoint: the toy data give CCEI** $= 0.875$**, exactly** $7/8$**.** Then: your own CCEI, and a loop over every ID in the file to build the class distribution — the histogram goes on the projector, with you marked on it.

### Part 4 — Would this test catch anyone? (˜15 min)

Before anyone brags: score 100 simulated *random* choosers — `runif(25, 0, 100)` tokens in H, converted to bundles exactly as the quiz was — with the identical code, on our actual prices (Bronars 1987). If random behavior scores well, a high CCEI means nothing. **Checkpoint: their mean CCEI is about** 0.68 **on our budgets** — so it doesn't, and yours does.

## Deliverable

Two pages, due **[date — suggest: Thursday 8 October, before EC1]**, submitted via **[method]**. Contents: (i) your CCEI and where it sits in the class distribution; (ii) the random-chooser figure with one paragraph interpreting it; (iii) one disciplined paragraph: *what does a CCEI of 1 actually* *certify, and what evidence would make you distrust a high score?* Consider the classmate who put all 100 tokens in the cheaper account every single round.

## References

Varian (1982), *Econometrica*, §1–2; Bronars (1987), *Econometrica*; Afriat (1967); Choi, Fisman, Gale & Kariv (2007), *AER*; Choi, Kariv, Müller & Silverman (2014), *AER*; Chen, Liu, Shan, Zhong & Zhou (2023), *PNAS*; Chambers & Echenique, *Revealed Preference Theory*, Ch. 3–4.